

σ -Algebra/Borel field	Kolmogorov's Axioms
Probability Measure Space	Bonferroni's Inequality
Law of Total Probability	Boole's Inequality
Event	DeMorgan's Laws
Disjoint/Mutually Exclusive Events	Exhaustive Events

Given a sample space $\mathcal{S}$ and an associated σ-algebra $\mathcal{B}$, a probability function is a function P with domain $\mathcal{B}$ that satisfies 1. $P(A) \geq 0 \quad \forall A \in \mathcal{B}$ 2. $P(\mathcal{S}) = 1$ 3. If $A_1, A_2, \dots \in \mathcal{B}$ are pairwise disjoint then $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$	A collection of subsets of $\mathcal{S}$ is called a σ-algebra or Borel field $\mathcal{B}$ if 1. $\emptyset \in \mathcal{B}$ 2. $A \in \mathcal{B} \implies A^c \in \mathcal{B}$ ie $\mathcal{B}$ is closed under complementation 3. If $A_1, A_2, \dots \in \mathcal{B}$ then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{B}$ ie $\mathcal{B}$ is closed under countable unions
$P(A \cap B) \geq P(A) + P(B) - 1$	If $\mathcal{B}$ is a σ-algebra field on $\mathcal{S}$ and P is a probability measure satisfying Kolmogorov's Axioms, then the triple $(\mathcal{S}, \mathcal{B}, P)$ is called a probability measure space.
For $(\mathcal{S}, \mathcal{B}, P)$ with $A \in \mathcal{B}$ $P(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} P(A_i) \text{ for any collection of sets } A_1, A_2, \dots \in \mathcal{B}$	For $(\mathcal{S}, \mathcal{B}, P)$ with $A \in \mathcal{B}$ $P(A) = \sum_{i=1}^{\infty} P(A \cap C_i) \text{ for } C_1, C_2, \dots \in \mathcal{B} \text{ where } C_1, C_2, \dots \text{ are a partition of } \mathcal{S}$
1. $(A \cup B)^c = A^c \cap B^c$ 2. $(A \cap B)^c = A^c \cup B^c$	Any collection of possible outcomes of an experiment; that is any subset of $\mathcal{S}$ 1. Simple event: Event with a single outcome 2. Compound event: Combination of two or more simple events
If $A_1, A_2, \dots$ are such that $\bigcup_{i=1}^{\infty} A_i = \mathcal{S}$ then the collection of events is said to be exhaustive.	Two events A and B defined on $\mathcal{S}$ are disjoint (mutually exclusive) if $A \cap B = \emptyset$. The events $A_1, A_2, \dots$ are pairwise disjoint if $A_i \cap A_j = \emptyset \quad \forall i \neq j$

Partition of S	Fundamental Theorem of Counting
Ordered with Replacement	Unordered without Replacement
Ordered without Replacement	Unordered with Replacement
Conditional Probability	Bayes Rule
Statistical Independence	Mutually Independent Events

If a job consists of k separate tasks, the i^{th} of which can be done in n_i ways, $i = 1, 2, \dots, k$, then the entire job can be done in $n_1 \times n_2 \times \dots \times n_k$ ways.	If $A_1, A_2, \dots$ are mutually exclusive and exhaustive, then they form a partition of $\mathcal{S}$
In a population with n unordered units, selecting r without replacement $\binom{n}{r} = \frac{n!}{(n-r)!r!}$	In a population with n ordered units, selecting r with replacement n^r
In a population with n unordered units, selecting r with replacement $\binom{n+r-1}{r} = \binom{n+r-1}{n-1} = \frac{(n+r-1)!}{(n-1)!r!}$	In a population with n ordered units, selecting r without replacement $\frac{n!}{(n-r)!}$
$P(A_i B) = \frac{P(B A_i)P(A_i)}{\sum_{j=1}^{\infty} P(B A_j)P(A_j)}$	$P(A B) = \frac{P(AB)}{P(B)}$
A collection of events $A_1, \dots, A_n$ are mutually independent if for any subcollection $A_{i1}, \dots, A_{ik}$ we have $P\left(\bigcap_{j=1}^k A_{ij}\right) = \prod_{j=1}^k P(A_{ij})$	The events A and B are independent if • $P(AB) = P(A \cap B) = P(A)P(B)$ OR • $P(A B) = P(A)$ OR • $P(B A) = P(B)$

Random Variable	Induced Probability Function
Cumulative Distribution Function (CDF)	Properties of a CDF
Identically Distributed	Probability Density Function (PDF)
Probability Mass Function (PMF)	Properties of PMF and PDF
PDF of a Transformation	Probability Integral Transform (PIT)

Consider a finite sample space $\mathcal{S} = \{\omega_1, \dots, \omega_n\}$ with probability function P. Define the random variable $X(\mathcal{S}) \mapsto \mathcal{X}$ where $\mathcal{X} = \{x_1, \dots, x_m\}$. Then $X = x_i$ if and only if the outcome is a $\omega_j \in \mathcal{S} : X(\omega_j) = x_i$. The induced probability function P_X on $\mathcal{X}$ is $P_X(X = x_i) = P(\{\omega_j \in \mathcal{S} : X(\omega_j) = x_i\})$	A random variable is a function from the sample space $\mathcal{S}$ into the real numbers $\mathbb{R}$ ie $X(\mathcal{S}) \mapsto \mathbb{R}$
$F_X(x)$ is a CDF iff $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow \infty} F_X(x) = 1$ $F_X(x)$ is a nondecreasing function of x $F_X(x)$ is right-continuous ie $\lim_{\epsilon \rightarrow 0^+} F_X(x + \epsilon) = F_X(x)$	$F_X(x) = P_X(X \leq x) \quad \forall x \in \mathcal{X}$
The PDF of a continuous random variable X is the function $f_X(x) \geq 0$ that satisfies $F_X(x) = \int_{-\infty}^x f_X(t) dt \quad \forall x \in \mathcal{X}$	Let $\mathcal{B}'$ be the smallest σ-algebra containing all intervals of real numbers of the form $(a, b), (a, b], [a, b), [a, b]$. The random variables X and Y are identically distributed for every set $A \in \mathcal{B}'$ if $P(X \in A) = P(Y \in A)$
A function $f_X(x)$ is a PDF or PMF of a random variable X iff $f_X(x) \geq 0 \quad x \in \mathcal{X}$ - If X is discrete then $\sum_{\text{all } x \in \mathcal{X}} f_X(x) = 1$. - If X is continuous then $\int_{-\infty}^{\infty} f_X(x) dx = 1$	The PMF of a discrete random variable X is the function $f_X(x) \geq 0$ that satisfies $F_X(x) = \sum_{t \leq x} f_X(t) \quad \forall x \in \mathcal{X}$ Also note that $f_X(x) = P(X = x)$
Let X have continuous CDF $F_X(x)$ and define the random variable Y as $Y = F_X(x)$. Then Y is uniformly distributed on $(0, 1)$; that is $P(Y \leq y) = y \text{ for } 0 < y < 1$	Let X have PDF $f_X(x)$ and $Y = g(X)$. $f_Y(y) = \sum_{i=1}^k f_X(g_i^{-1}(y)) \left \frac{\partial}{\partial y} g_i^{-1}(y) \right \quad \forall y \in \mathcal{Y}$ (See Slide 24 of <code>distfrv(transformations)</code> part 2 for necessary conditions)

Expected Value $\mathbb{E}[X]$	Law of the Unconscious Statistician (LOTUS)
Moments and Central Moments of a Distribution	Variance of a Distribution
Moment Generating Function (MGF)	MGF of a Linear Transformation
Leibnitz's Rule	Binomial Theorem
Geometric Series and Properties	Memoryless Property

1. If X is discrete with PMF $f_X(x)$ then $\mathbb{E}[g(X)] = \sum_{x: f_X(x) > 0} g(x) f_X(x)$ 2. If X is continuous with PDF $f_X(x)$ then $\mathbb{E}[g(X)] = \int_{x: f_X(x) > 0} g(x) f_X(x) dx$	Let X be a random variable. Then $\mathbb{E}[X]$ is 1. If X is discrete with PMF $f_X(x)$ then $\mathbb{E}[X] = \sum_{x: f_X(x) > 0} x f_X(x)$ 2. If X is continuous with PDF $f_X(x)$ then $\mathbb{E}[X] = \int_{x: f_X(x) > 0} x f_X(x) dx$
$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$	- For each integer n, the n^{th} moment of X (or of $F_X(x)$) denoted μ'_n is defined as $\mu'_n = \mathbb{E}[X^n]$ - The n^{th} central moment of X (or of $F_X(x)$) denoted μ_n is defined as $\mu_n = \mathbb{E}[(X - \mu)^n]$
$M_{aX+b}(t) = e^{bt} M_X(at)$	$M_X(t) = \mathbb{E}[e^{tX}]$ (Also look at slides 29 and 30 of moments)
$(a+b)^r = \sum_{j=0}^r \binom{r}{j} a^j b^{r-j}$	$\begin{aligned} \frac{\partial}{\partial \theta} \int_{a(\theta)}^{b(\theta)} f(x, \theta) dx = \\ f(b(\theta), \theta) \frac{\partial}{\partial \theta} b(\theta) - \\ f(a(\theta), \theta) \frac{\partial}{\partial \theta} a(\theta) + \int_{a(\theta)}^{b(\theta)} \frac{\partial}{\partial \theta} f(x, \theta) dx \end{aligned}$
Let s and t be two integers such that $s > t$. The distribution is said to be memoryless if $P(X > s X > t) = P(X > s - t)$ OR $P(X > m + n X > m) = P(X > n)$	Geometric series: $\sum_{k=0}^{\infty} ar^k$. If $r < 1$ then $\sum_{k=0}^{m-1} ar^k = a \left(\frac{1-r^m}{1-r} \right)$ $\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}$

Gamma Function	Beta Function
Exponential Families	Indicator Function
Properties of Exponential Family	Natural Parameter Space
Curved Exponential Family	Location-Scale Family
Location-Scale Families: Mean and Variance	Joint PMF

Let $\alpha > 0$ and $\beta > 0$. Then $B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$	Let $z > 0$. Then $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$ 1. $\Gamma(z+1) = z\Gamma(z)$ 2. If $z \in \mathbb{Z}^+$ then $\Gamma(z) = (z-1)!$ 3. $\Gamma(0.5) = \sqrt{\pi}$ 4. Legendre duplication formula: $\Gamma(z)\Gamma(z+0.5) = 2^{1-2z}\sqrt{\pi}\Gamma(2z)$
$I_A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$	A PDF/PMF is an exponential family if its can be expressed as $f(x \boldsymbol{\theta}) = h(x) c(\boldsymbol{\theta}) \exp \left\{ \sum_{j=1}^k w_j(\boldsymbol{\theta}) t_j(x) \right\}$ • $h(x) \geq 0$ and $c(\boldsymbol{\theta}) \geq 0$ • $t_1(x), \dots, t_k(x)$ are real-valued functions of x • $w_1(\boldsymbol{\theta}), \dots, w_k(\boldsymbol{\theta})$ are real-valued functions of $\boldsymbol{\theta}$
The exponential family can be reparameterized as $f(x \boldsymbol{\eta}) = h(x) c^*(\boldsymbol{\eta}) \exp \left\{ \sum_{i=1}^k \eta_i t_i(x) \right\}$ where the set $\mathcal{H} = \left\{ \boldsymbol{\eta} : \int_{-\infty}^{\infty} h(x) \exp \left\{ \sum_{i=1}^k \eta_i t_i(x) \right\} dx < \infty \right\}$ is the natural parameter space of the family.	1. $\mathbb{E} \left[\sum_{i=1}^k \frac{\partial w_i(\boldsymbol{\theta})}{\partial \theta_j} t_i(X) \right] = -\frac{\partial}{\partial \theta_j} \log c(\boldsymbol{\theta})$ 2. $\text{Var} \left(\sum_{i=1}^k \frac{\partial w_i(\boldsymbol{\theta})}{\partial \theta_j} t_i(X) \right) = -\frac{\partial^2}{\partial \theta_j^2} \log c(\boldsymbol{\theta}) - \mathbb{E} \left[\sum_{i=1}^k \frac{\partial^2 w_i(\boldsymbol{\theta})}{\partial \theta_j^2} t_i(X) \right]$
Let $f_X(x)$ be any PDF. Then for any $\mu, -\infty < \mu < \infty$, and any $\sigma > 0$, the family of PDFs $\frac{1}{\sigma} f((x-\mu)/\sigma)$ is the location-scale family with PDF $f(x)$ where μ is the location parameter of σ is the scale parameter.	A curved exponential family is a family of densities of the form of the exponential family for which the dimension of the vector $\boldsymbol{\theta}$ is $d < k$. If $d = k$, then the family is a full exponential family.
$f_{X,Y}(x, y) = P(X = x, Y = y)$	Let Z be a random variable with PDF $f_Z(z)$. Suppose $\mathbb{E}[Z]$ and $\text{Var}(Z)$ exists. If X is a random variable with PDF $\frac{1}{\sigma} f((x-\mu)/\sigma)$ then • $\mathbb{E}[X] = \sigma \mathbb{E}[Z] + \mu$ • $\text{Var}(X) = \sigma^2 \text{Var}(Z)$

Marginal PMF	Joint PDF
Marginal PDF	Joint CDF
Conditonal Distribution	Conditional Expectations
Independent Random Variables	Expectation of Independent Random Variables
MGF of $X + Y$ for Independent Random Variables	NOTE

A function $f_{X,Y}(x, y)$ is the joint PDF if for every $A \in \mathbb{R}^2$ $P((X, Y) \in A) = \int \int_A f_{X,Y}(x, y) dx dy$	$f_X(x) = \sum_{y \in \mathbb{R}} f_{X,Y}(x, y)$ $f_Y(y) = \sum_{x \in \mathbb{R}} f_{X,Y}(x, y)$
$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$	$f_X(x) = \int_{y \in \mathbb{R}} f_{X,Y}(x, y) dy$ $f_Y(y) = \int_{x \in \mathbb{R}} f_{X,Y}(x, y) dx$
- For discrete $\mathbb{E}[g(Y) x] = \sum_y g(y) f(y x)$ - For continuous $\mathbb{E}[g(Y) x] = \int_y g(y) f(y x) dy$	$f_{Y X}(y x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$ $f_{X Y}(x y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$
Let X and Y be independent random variables and $g(x)$ be a function of x alone and $h(y)$ be a function of y alone. Then $\mathbb{E}[g(X) h(Y)] = \mathbb{E}[g(X)] \mathbb{E}[h(Y)]$	X and Y are independent random variables if for every $x \in \mathbb{R}$ and $y \in \mathbb{R}$ $f_{X,Y}(x, y) = f_X(x) f_Y(y)$
KNOW HOW TO DO BIVARIATE TRANSFORMATIONS!!!	Let $X \perp Y$ with MGFs $M_X(t)$ and $M_Y(t)$. Then the MGF of $S = X + Y$ is given by $M_S(t) = M_X(t) M_Y(t)$

Hierarchical Expectations	Mixture Distribution
Conditional Variance Identity	Covariance
Correlation	Variance of Linear Combination of X and Y
Properties of Bivariate Normal Distribution	Sample Mean
Sample Variance	Expectations and Variances of Functions

A random variable X is said to have a mixture distribution if the distribution of X depends on a quantity that also has distribution	If X and Y are two random variables then $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X Y]]$
$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_x)(Y - \mu_Y)] = \mathbb{E}[XY] - \mu_X \mu_Y$	$\text{Var}(X) = \mathbb{E}[\text{Var}(X Y)] + \text{Var}(\mathbb{E}[X Y])$
$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$ If $X \perp Y$ then $\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$	$\rho_{X,Y} = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$
$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$	1. $f_X(x) \sim N(\mu_X, \sigma_X^2)$ 2. $f_Y(y) \sim N(\mu_Y, \sigma_Y^2)$ 3. $\rho_{XY} = \rho$ 4. The distribution of $aX + bY$ is normal with mean $a\mu_X + b\mu_Y$ and variance $a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\rho\sigma_X\sigma_Y$
If $X_1, \dots, X_n$ are <i>iid</i> then • $\mathbb{E}[\sum_{i=1}^n g(X_i)] = n\mathbb{E}[g(X_1)]$ • $\text{Var}(\sum_{i=1}^n g(X_i)) = n\text{Var}(g(X_1))$	$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$

Properties of $\bar{X}$ and S^2	Convolution Formula (Continuous)
Properties of $\bar{X}$ and S^2 (Part 2)	χ^2 Distribution Properties
Snedecor's F distribution	Properties of the F distribution
j^{th} Order Stat PDF	PDF of $X_{(1)}$
PDF of $X_{(n)}$	Convergence in Probability

If $X \perp Y$ with PDFs $f_X(x)$ and $f_Y(y)$ then the PDF of $W = X + Y$ is $f_W(w) = \int_{-\infty}^{\infty} f_X(t)f_Y(w-t)dt$	If $X_1, \dots, X_n$ are <i>iid</i> with mean μ and variance $\sigma^2 < \infty$ then 1. $\mathbb{E}[\bar{X}] = \mu$ 2. $\text{Var}(\bar{X}) = \sigma^2/n$ 3. $\mathbb{E}[S^2] = \sigma^2$
1. If $Z \sim N(0, 1)$ then $Z^2 \sim \chi_1^2$ 2. If $X_1, \dots, X_n$ are independent with $X_i \sim \chi_{\nu_i}^2, i = 1, \dots, n$ then $\sum_{i=1}^n X_i \sim \chi_v^2$ where $v = \sum_{i=1}^n \nu_i$	Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$ and let $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$. Then 1. $\bar{X} \perp S^2$ 2. $\bar{X} \sim N(\mu, \sigma^2/n)$ 3. $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{(n-1)}^2$
1. If $X \sim t_\nu$ then $X \sim F_{1,\nu}$ 2. If $X \sim F_{\nu_1, \nu_2}$ then $X^{-1} \sim F_{\nu_2, \nu_1}$ 3. If $X \sim F_{\nu_1, \nu_2}$ and $Y = (\nu_1/\nu_2)X$ then $Y/(1+Y) \sim \text{Beta}(\nu_1/2, \nu_2/2)$	If $X \perp Y$ and $X \sim \chi_{\nu_1}^2$ and $Y \sim \chi_{\nu_2}^2$ then $F = \frac{X/\nu_1}{Y/\nu_2} \sim F(\nu_1, \nu_2)$
$f_{X_{(1)}}(x) = n f_X(x) (1 - F_X(x))^{n-1}$	$f_{X_{(j)}}(x) = \frac{n!}{(j-1)!(n-j)!} f_X(x) F_X(x)^{j-1} (1 - F_X(x))^{n-j}$
A sequence of random variables $X_1, \dots, X_n$ converges in probability to a random variable X if for every $\epsilon > 0$ • $\lim_{n \rightarrow \infty} P(X_n - X \geq \epsilon) = 0$ OR • $\lim_{n \rightarrow \infty} P(X_n - X < \epsilon) = 1$	$f_{X_{(n)}}(x) = n f_X(x) F_X(x)^{n-1}$

Weak Law of Large Numbers (WLLN)	Consistent Estimator
Convergence in Distribution	Central Limit Theorem (CLT)
Slutzky's Theorem	Taylor's Polynomial of Order r
Taylor's Theorem	First-Order Delta Method
Second-Order Delta Method	

Consider a statistic $T_n(X_1, \dots, X_n)$. If $T_n \xrightarrow{p} \theta$ as $n \rightarrow \infty$ then T_n is a consistent estimator of θ.	$\bar{X}_n \xrightarrow{p} \mu$
$\sqrt{n} (\bar{X} - \mu) \xrightarrow{d} N(0, \sigma^2)$	A sequence of random variables $X_1, \dots, X_n$ converges in distribution to a random variable X if $\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$ at all points x where $F_X(x)$ is continuous
If a function $g(x)$ has derivatives of order r that is $g^{(r)}(x) = \frac{\partial^r}{\partial x^r} g(x)$ exists, then for any constant a, the Taylor polynomial of order r about a is $T_r(x) = \sum_{j=0}^r \frac{g^{(j)}(a)}{j!} (x-a)^j$	If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} a$ where a is a real constant, then 1. $Y_n X_n \xrightarrow{d} aX$ 2. $X_n + Y_n \xrightarrow{d} X + a$
Let Y_n be a sequence of random variables that satisfies $\sqrt{n}(Y_n - \theta) \xrightarrow{d} N(0, \sigma^2)$. For a given function g and a specific value of θ, suppose that $g'(\theta) \neq 0$ and exists. Then $\sqrt{n}(g(Y_n) - g(\theta)) \xrightarrow{d} N(0, \sigma^2 g'(\theta)^2)$	If $g^{(r)}(a) = \frac{\partial^r}{\partial x^r} g(x) _{x=a}$ exists, then $\lim_{x \rightarrow a} \frac{g(x) - \text{Tr}(x)}{(x-a)^r} = 0 \text{ where}$ $g(x) - \text{Tr}(x) = \int_a^x \frac{g^{(r)}(t)}{r!} (x-t)^r dt$
	Let Y_n be a sequence of random variables that satisfies $\sqrt{n}(Y_n - \theta) \xrightarrow{d} N(0, \sigma^2)$. For a given function g and a specific value of θ, suppose that $g'(\theta) = 0$ and $g''(\theta) \neq 0$ and exists. Then $n(g(Y_n) - g(\theta)) \xrightarrow{d} \frac{1}{2} \sigma^2 g''(\theta) \chi_1^2$